

# Efficient Fine-grained Image Retrieval with Vision Foundation Models for Industrial Objects

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## Abstract

Image-based object retrieval in industrial scenarios remains highly challenging due to the combination of fine-grained visual similarity between parts and significant appearance variations caused by viewpoint, background, and illumination changes. In this work, we introduce a large-scale dataset for fine-grained instance-level retrieval of industrial spare parts, comprising more than 17k object instances captured under both controlled and mobile acquisition conditions. Designed to reflect realistic retrieval challenges, the dataset provides a standardized benchmark for fine-grained industrial matching. Using this benchmark, we systematically compare three recent vision foundation models—DINOv2, DINOv3, and SigLIP2—analyze how token representations affect performance, with a particular focus on the roles of CLS and patch tokens and on pooling strategies. To bridge the gap between generic pretrained representations and retrieval-specific requirements, we utilize a lightweight framework that trains on top of frozen encoders. Results across multiple evaluation settings demonstrate consistent gains over off-the-shelf embeddings and suggest that patch-token representations can play an important role in fine-grained image retrieval, especially under larger viewpoint and background variations. The dataset is publicly available at: [https://github.com/imyiliu1717/industrial\\_spare\\_part\\_dataset](https://github.com/imyiliu1717/industrial_spare_part_dataset).

## 1. INTRODUCTION

Instance-level object retrieval aims to identify the correct object in a large image database given a query image. This capability is particularly important in industrial applications (e.g., maintenance, warehouse logistics, inspection, or

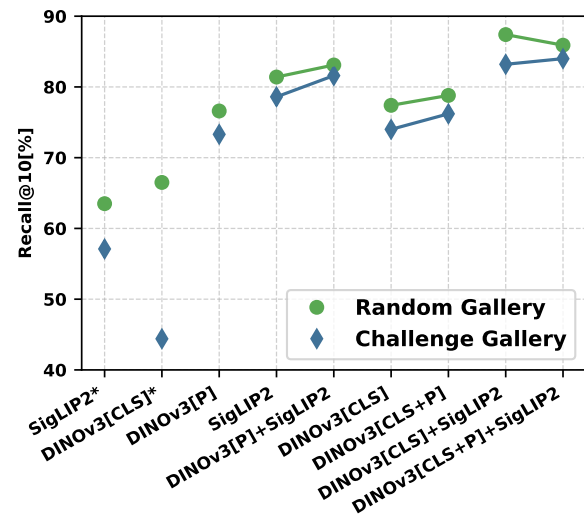


Figure 1. **Image retrieval performance** on Random and Challenge spare part object image gallery using vision foundation model embeddings. Star \* indicate the pre-train model. Each embedding is represented by two points: Only CLS token and with patch tokens (P). Lines connect the points, highlighting the improvement in retrieval performance when patch tokens are incorporated.

robotic picking), where rapidly matching a query against large reference collections of industrial spare parts can improve the efficiency of inspection, categorization, maintenance, and other downstream workflows. At the same time, it is especially challenging because many parts differ only in subtle geometric properties (Fig. 2), such as small variations in length, shape, or structure, which may be barely distinguishable in the image space. These difficulties are further amplified by viewpoint changes, specular surfaces, motion blur, and cluttered backgrounds. Moreover, many spare

parts are difficult to describe unambiguously in natural language, making image-based retrieval a practical and often necessary interface. Despite substantial progress in instance retrieval for landmarks and consumer products, standardized large-scale benchmarks for industrial retrieval under realistic acquisition variability remain limited.

Recent vision foundation models provide powerful transferable representations, but they are not necessarily optimized for distinguishing fine-grained industrial instances. In particular, Vision Transformers expose both a global CLS token and spatial patch tokens, and it remains unclear which signals are most reliable for industrial retrieval and under what conditions. Patch tokens may retain localized geometric cues, but naïve aggregation can amplify background noise. Additionally, different foundation models are trained with different objectives (self-supervised vs. vision-language), suggesting that their embeddings may be complementary—if they can be combined in a retrieval-friendly way.

In this work, we address these gaps with a new benchmark and a lightweight adaptation strategy that improves retrieval while keeping the foundation encoders frozen.

We summarize our contributions as follows:

- **Benchmarking on a new Industrial Spare Parts dataset:** a large-scale multi-view benchmark for industrial instance retrieval (17k+ objects) with controlled and mobile acquisition conditions and retrieval-oriented splits. The dataset is currently undergoing an internal approval process. Subject to successful completion of this review, it is planned for public release in due course.
- **Encoder study:** a thorough comparison of DINOv2 [22], DINOv3 [31], and SigLIP2 [36] for industrial retrieval, including robustness-focused evaluation settings using a lightweight adaptation framework.
- **Token-level analysis:** an investigation of how CLS vs. patch tokens and different pooling strategies influence retrieval performance and robustness (especially under larger viewpoint gaps).

## 2. RELATED WORK

### 2.1. Instance-Level Retrieval

Early instance-level image retrieval relied on interest point detectors and hand-crafted descriptors with bag-of-visual-words indexing and spatial verification for scalable retrieval [18, 19, 24, 32]. Later, local descriptors were aggregated into global representations such as VLAD and Fisher vectors, enabling compact and effective large-scale retrieval with efficient nearest-neighbor search [14, 23].

With deep learning, retrieval shifted to CNN-based descriptors and end-to-end embedding learning; globally pooled and fine-tuned representations became standard baselines [2, 9]. Deep local and region-based methods im-

proved robustness: DELF and DELG emphasized salient regions and unified local-global signals [4, 20], while R-MAC, NetVLAD, and related aggregation schemes preserved local evidence within compact global embeddings for scalable retrieval [1, 34].

Projection heads or supervised fine-tuning are commonly used to adapt feature embeddings to fine-grained retrieval tasks [15, 26]. Similarly, attention mechanisms and dedicated CNN architectures focusing on object sub-parts have been studied to improve performance on fine-grained visual categorization tasks [39].

Many large-scale systems follow a two-stage design: a fast first-pass nearest-neighbor search with compact global descriptors produces a shortlist, which is then refined using more expensive local matching and geometric verification (often RANSAC-based spatial verification) [1, 8, 24, 34]. Recent transformer-based methods learn the re-ranking function directly over sets of local descriptors/tokens; R2Former unifies retrieval and re-ranking within a single transformer architecture [43]. AMES further improves scalability by performing asymmetric, memory-efficient similarity estimation, allowing rich query descriptors while storing compact database representations [33].

### 2.2. Vision Foundation Models for Image Retrieval

Modern retrieval increasingly builds on large pretrained encoders. Vision Transformers expose a CLS token alongside patch tokens [7]. Self-supervised ViTs (DINO) and related pretraining methods (MoCo v3, MAE) yield transferable token features [5, 6, 12], while vision-language models provide strong semantic similarity spaces (CLIP, SigLIP) [27, 36, 41]. CaiT and TokenLearner showed that interactions between CLS and patch-level tokens can improve embeddings [29, 35].

### 2.3. Benchmarks & Datasets

Common benchmarks cover landmarks (Revisited Oxford/Paris; GLDV2) [25, 38] and consumer products/fashion (SOP, DeepFashion, eProduct, Product1M) [10, 17, 21, 40, 42]. At large scale, ILIAS evaluates retrieval with a massive 100-million-image distractor set [16]. While ILIAS is explicitly curated to test robustness against variations in viewpoint, scale, and occlusions, these datasets typically do not include quantified view variations. In contrast to consumer-product-focused datasets, widely used industrial datasets mainly target other tasks (e.g., anomaly detection or pose estimation, such as MVTec AD and T-LESS) [3, 13], leaving limited standardized coverage for large-scale industrial instance retrieval.



Figure 2. **Object Examples.** Each row shows different objects from the same category: small metal parts with subtle structural differences, springs of different lengths, cables with different connectors, and PCBs with different layouts.



Figure 3. **Representative examples** from the query set and the three gallery subsets (random, challenge, and machine-image).

### 3. Spare Parts Dataset

#### 3.1. Composition and Collection

We benchmark on a new dataset with 17,410 different kinds of industrial spare parts. The dataset contains two sets of images captured under different acquisition conditions. In the first setting (*machine-image gallery*), objects are captured by a high-quality camera from multiple viewpoints, under controlled lighting conditions and with a clean white background. In the second setting, objects are captured using a mobile phone from multiple arbitrary viewpoints, but with at most seven different backgrounds and uncontrolled illumination conditions. As a result, images in the second setting may exhibit motion blur and other quality degradations. All images contain a single object at the center without a cluttered background.

Table 1. Statistical overview of the dataset, including number of objects, images, number of images per object (Img/Obj), and capture diversity including number of backgrounds (Bg) per object and viewing directions per object.

| Image Set                    | Objects | Images  | Img/Obj | Bg/Obj | View/Obj |
|------------------------------|---------|---------|---------|--------|----------|
| <b>Mobile Phone Images</b>   | 17,410  | 131,937 | 7.57    | 4.59   | 4.32     |
| Train                        | 7,395   | 67,844  | 9.17    | 5.01   | 4.70     |
| Validation                   | 15      | 461     | 30.7    | 6.20   | 7.40     |
| Test                         | 9,526   | 37,520  | 3.94    | 3.00   | 2.88     |
| Query Images                 | 10,000  | 10,000  | 1.0     | 1.0    | 1.0      |
| Random Gallery               | 10,000  | 10,000  | 1.0     | 1.0    | 1.0      |
| Challenging Gallery          | 10,000  | 10,000  | 1.0     | 1.0    | 1.0      |
| <b>Machine-Image Gallery</b> | 10,000  | 10,000  | 1.0     | 1.0    | 1.0      |

#### 3.1.1. Large Variety of Objects

Our dataset presents several intrinsic challenges that make both recognition and retrieval tasks difficult. First, many objects are visually very similar, differing only in subtle attributes such as size, length, or small structural details. Even human experts may struggle to distinguish these objects at a glance. Second, the differences between objects are often hard to describe in natural language, as many instances are highly specific to industrial settings. For example, differentiating two printed circuit boards (PCBs) may require lengthy technical descriptions or domain-specific prior knowledge. Finally, most objects in our dataset are rarely seen in standard pretraining datasets, providing limited prior knowledge for models and requiring them to learn discriminative representations from scratch. Taken together, these factors create a challenging benchmark that tests models’ ability to capture subtle visual differences, leverage domain-specific knowledge, and generalize to unfamiliar object instances. Examples are presented in Figure 2.

#### 3.1.2. Statistics

We construct a large-scale dataset containing 17,410 objects and 141,937 images, including 131,937 mobile phone images and 10,000 machine images. The objects are divided into 7,395 / 15 / 10,000 objects for training, validation, and testing, respectively, ensuring strict identity separation across splits. Each object is captured with an average of five different backgrounds and four distinct viewing directions, providing substantial environmental and viewpoint diversity. This systematic acquisition allows controlled analysis of representation robustness while maintaining realistic appearance variations. In addition to the mobile phone imagery, we include 10,000 machine images. This subset introduces a natural domain shift in appearance statistics and capture conditions, enabling cross-domain evaluation. Detailed dataset statistics are summarized in Table 1.

## 3.2. Benchmarks

### 3.2.1. Image Identification

Industrial parts often exhibit strong structural similarity, making instance discrimination particularly challenging. To evaluate this setting, we formulate a binary image identification task in which the model must determine whether two images depict the same object instance. Since the dataset focuses on industrial parts with visually similar structures, we provide training data to train linear adaptations of different embeddings. We construct 40k / 4k / 20k mobile phone image pairs for training, validation, and testing, respectively. Each split contains an equal number of positive and negative pairs. Image pairs are randomly sampled while explicitly enforcing object diversity, ensuring broad coverage of instances and preventing memorization of specific objects. The validation split is intentionally smaller but denser in observations per object, providing stable signals for hyperparameter tuning. This benchmark focuses on learning discriminative embeddings and testing their ability to generalize to unseen object identities.

### 3.2.2. Image Retrieval

We introduce an image retrieval benchmark to evaluate large-scale instance search and robustness to appearance variation. All query and gallery images are sampled exclusively from test object identities to ensure strict separation from training data, with each query having exactly one valid match. For each object, two gallery images are used to form two sets: a **Random Gallery**, containing a randomly sampled image per object for standard retrieval, and a **Challenge Gallery**, containing an image differing from the query in background and viewpoint to test robustness. To evaluate cross-domain generalization, we further build a Machine-Image Gallery by sampling one machine-captured image per object, introducing a realistic domain shift. Example images are shown in Figure 3.

### 3.2.3. Metrics

For the binary identification task, we report accuracy, which reflects overall classification performance. Since each query has exactly one relevant gallery image, we evaluate retrieval performance using Recall@k and Mean Reciprocal Rank (MRR@k), focusing on top-k results with  $k \in \{1, 10, 100\}$  to align with industrial deployment requirements. **Recall@k** measures the fraction of queries for which the correct match appears among the top-k retrieved results:

$$\text{Recall@k} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(r_i \leq k), \quad (1)$$

where  $N$  is the number of queries,  $r_i$  is the rank of the correct match for query  $i$ , and  $\mathbf{1}(\cdot)$  is the indicator function. **MRR@k** evaluates the ranking quality by considering the

position of the first correct match while ignoring ranks beyond the cutoff  $k$ :

$$\text{MRR@k} = \frac{1}{N} \sum_{i=1}^N \begin{cases} \frac{1}{r_i}, & r_i \leq k \\ 0, & r_i > k \end{cases}. \quad (2)$$

Under this single-positive setting, the Average Precision (AP) for each query reduces to the reciprocal rank of the relevant item; therefore, MRR@k is equivalent to the truncated mean Average Precision (mAP@k) in our case. Both metrics are reported for the Random, Challenging, and Machine-image gallery sets. This truncated metric reflects the practical observation that only the top-ranked candidates are inspected in industrial deployments, emphasizing performance within an operationally relevant retrieval range.

## 4. Fine-grained Image Retrieval

### 4.1. Problem Statement

Given a target spare part, our objective is to identify the corresponding object instance within a large-scale object database. We formulate this problem as an image-based object retrieval task. The database contains multiple images of each object captured under diverse viewing conditions, including variations in background, viewpoint, and illumination. Given a query image of the target object, the objective is to retrieve from the database the image corresponding to the same physical object instance by ranking candidate images according to a similarity score.

Formally, let  $G = I_1, \dots, I_N$  be a gallery of images with instance labels  $y_i = L(I_i)$ . For a query image  $I_q$  with label  $y_q$ , we aim to produce a ranking of  $G$  based on similarity to  $I_q$ . This is achieved by learning:

1. An embedding function  $E : \mathcal{I} \rightarrow \mathbb{R}^d$  that maps an image  $I$  from the image space  $\mathcal{I}$  to a  $d$ -dimensional embedding vector.
2. A similarity function  $S : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, 1]$  that computes a scalar similarity score between two embeddings.

The optimal functions  $E$  and  $S$  should satisfy the property that for any two images  $I_a$  and  $I_b$ :

$$S(E(I_a), E(I_b)) \rightarrow 1 \quad \text{if } L(I_a) = L(I_b) \quad (3)$$

$$S(E(I_a), E(I_b)) \rightarrow 0 \quad \text{if } L(I_a) \neq L(I_b) \quad (4)$$

For any query  $I_q$ , the gallery  $G$  is then ranked by sorting images  $I_i$  in descending order of their computed similarity scores  $s_i = S(E(I_q), E(I_i))$ . We train the model using a pairwise binary classification objective to compute the similarity score. Given a query-gallery image pair, the network predicts whether the two images correspond to the same object. This formulation allows the model to learn a task-specific similarity function. At inference time, the predicted matching probability (or logit) is used as a similarity score to rank all gallery images for a given query.



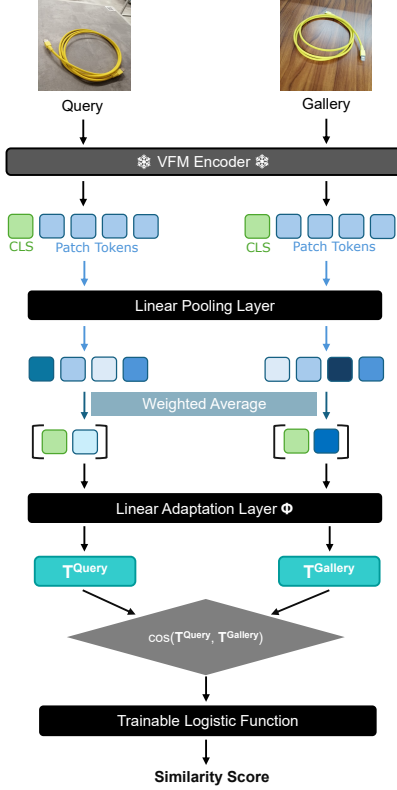


Figure 4. Architecture.

## 4.2. Architecture Overview

Our similarity scoring network consists of a frozen vision foundation model used as an image encoder, followed by a lightweight token aggregation and projection module trained for instance-level representation learning, and a trainable discrimination function. Given an input image, the foundation encoder produces a global CLS token and a set of local patch tokens. The proposed aggregation model assigns importance weights to patch tokens, aggregates them into a compact object-level representation, and fuses this representation with the global CLS token to obtain the final embedding used for similarity computation by the discrimination function. This design, as visualized in Figure 4, allows us to leverage powerful, pre-trained visual features while efficiently adapting them to the specific challenge of fine-grained, instance-level discrimination.

### 4.2.1. Encoder: Vision Foundation Models

The overall performance of the retrieval pipeline is highly dependent on the choice of the vision backbone. Foundation models that are trained on vast image datasets provide a strong prior for comparing similarities. In this work, we experiment with image embeddings from three state-of-the-art vision (-language) foundation models: DINOv2 [22], DINOv3 [31], and SigLIP2 [36]. These models output CLS

tokens that capture global semantic information, as well as patch tokens to preserve richer local and geometric details. A straightforward way of comparing similarities is to directly compare CLS tokens, which serves as a baseline for our further experiments. See Table 4 for a comparison of the used foundation models. We also explore different pooling strategies for utilizing patch tokens and combining features from multiple foundation models, as described next.

### 4.2.2. Aggregation and Linear Adaptation

While the CLS token primarily captures global semantic information, patch tokens preserve richer local and geometric details. We therefore combine the CLS token  $T^{CLS}$  with  $N_P$  patch tokens  $\{T_i^P\}_{i=1}^{N_P}$  to form a more expressive representation. Since they have different dimensions, we first aggregate patch tokens with  $C_P$  channels over a batch  $B$  to a tensor of shape  $B \times C_P$  and then concatenate it with  $T^{CLS}$ . A straightforward way is to take the average over patches (Average Pooling (AP)). However, it will introduce background patches which could be irrelevant to the object. An alternative is to take the weighted average over patches (Weighted-average Pooling (WP)). Given the patch token tensor of shape  $B \times N_P \times C_P$ , where  $N_P$  denotes the number of patches, all patch tokens are passed through a shared linear layer  $F : \mathbb{R}^{C_P} \rightarrow \mathbb{R}$  to compute a scalar score for each patch. These scores are normalized across patches using a softmax function to obtain the patch weights  $\{w_i\}$ , ensuring that  $\sum_{i=1}^{N_P} w_i = 1$ . We observe that this attention-like weighting mechanism encourages the representation to focus on object-relevant regions while suppressing background features, without applying segmentation pre-processing, as illustrated in Figure 5. The patch tokens are then aggregated via a weighted average, yielding a single patch-level representation  $T^P$  of shape  $B \times C$ , where

$$T^P = \sum_i^{N_P} w_i T_i^P, \quad (5)$$

that is dimensionally aligned with the CLS token of shape  $B \times C^{CLS}$ . The final representation  $T^I$  with dimension  $C$  of an image  $I$  is computed by passing the concatenated tokens of  $T^{CLS}$  and  $T^P$  through a linear projection head  $\Phi : \mathbb{R}^{C^{CLS}+C_P} \rightarrow \mathbb{R}^C$ ,

$$T^I = \Phi(\text{concat}(T^{CLS}, T^P)). \quad (6)$$

Next, we need a distance measure to compare representations of two images.

### 4.2.3. Discrimination Function

To compute the similarity score of two images  $I_1, I_2$ , we first compute the cosine similarity between  $T^{I_1}, T^{I_2}$ .

However, different foundation models produce embeddings whose distance distributions vary significantly. This is partly because representations from contrastively trained

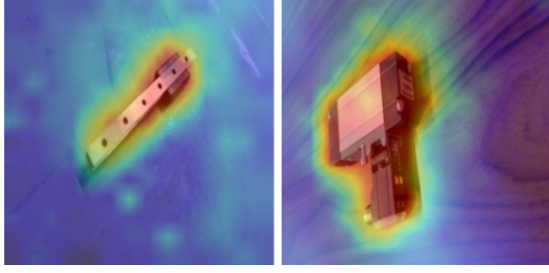


Figure 5. Attention weight maps of two example images.

models are known to be anisotropically distributed, collapsing into a narrow cone in the embedding space [30]. To account for this discrepancy, we normalize the cosine similarity score using a model-specific logistic function. Thus, the final similarity score  $S$  of two images  $I_1, I_2$  is computed by

$$S(I_1, I_2) = \frac{1}{1 + \exp(-(\mathbf{w}(1 - \cos(T^{I_1}, T^{I_2})) + \mathbf{b}))}, \quad (7)$$

where  $\mathbf{w}, \mathbf{b}$  are trainable scalar parameters.

### 4.3. Training

We use three pretrained Vision Transformer backbones as encoders: SigLIP2-SO400M/16-512, DINOv3-ViT-L/16 (LVD-1689M), and DINOv2-Large. All encoders are frozen, and we train only the linear pooling layer and linear projection head. The final output feature dimension  $C$  is 2048. We formulate image retrieval as a binary classification problem. The classifier is trained using binary cross-entropy loss. We use the AdamW optimizer with a learning rate of 0.001 and weight decay of 0.01. Models are trained for 64 epochs with a cosine learning rate schedule and a batch size of 64. Each configuration is trained with eight random seeds, and the best validation checkpoint is used for retrieval. We initialize the linear pooling layer using a truncated normal distribution with a standard deviation of 0.02 and zero bias. For the linear adaptation layer, we utilized the standard Kaiming uniform initialization [11].

## 5. Experiments

### 5.1. Image Identification

To evaluate the discriminative quality of visual features extracted from different pretrained encoders, we perform image identification on the dataset. Similarity scores are normalized using a fitted logistic function to ensure comparable decision thresholds across embedding types. The average accuracy and standard deviation across all runs of each embedding is reported in Table 2.

As shown in Table 2, pretrained global features (CLS tokens) already provide strong ability of object-level identification, achieving accuracies above 90%. In contrast,

Table 2. Mean and standard deviation (std) of image identification accuracy on 20k spare-part image pairs over eight seeds. A/WP denotes average/weighted-average pooling. All similarity scores are linearly normalized by logistic regression.

| Train                         | Embeddings           | Accuracy (%) ( $\pm$ std)      |                  |
|-------------------------------|----------------------|--------------------------------|------------------|
| Pre-trained                   | DINOv2[CLS]          | 91.33                          |                  |
|                               | DINOv3[CLS]          | 94.67                          |                  |
|                               | SigLIP2              | 94.26                          |                  |
|                               | DINOv2[P] + AP       | 72.77                          |                  |
|                               | DINOv3[P] + AP       | 62.33                          |                  |
|                               | DINOv2[CLS + P] + AP | 89.36                          |                  |
|                               | DINOv3[CLS + P] + AP | 93.63                          |                  |
| Linear Adaptation and Pooling | Global Features      | SigLIP2                        | 97.72 $\pm$ 0.29 |
|                               |                      | DINOv2[CLS]                    | 96.66 $\pm$ 0.37 |
|                               |                      | DINOv3[CLS]                    | 97.59 $\pm$ 0.31 |
|                               |                      | DINOv2[CLS] + SigLIP2          | 97.02 $\pm$ 0.65 |
|                               |                      | DINOv3[CLS] + SigLIP2          | 97.39 $\pm$ 0.66 |
|                               | +Local Features      | DINOv2[P] + WP                 | 96.54 $\pm$ 0.97 |
|                               |                      | DINOv3[P] + WP                 | 97.37 $\pm$ 0.40 |
|                               |                      | DINOv2[P] + SigLIP2 + WP       | 96.67 $\pm$ 0.56 |
|                               |                      | DINOv3[P] + SigLIP2 + WP       | 97.45 $\pm$ 0.61 |
|                               |                      | DINOv2[CLS + P] + WP           | 96.93 $\pm$ 0.73 |
|                               |                      | DINOv3[CLS + P] + WP           | 97.33 $\pm$ 0.61 |
|                               |                      | DINOv2[CLS + P] + SigLIP2 + WP | 97.03 $\pm$ 0.76 |
|                               |                      | DINOv3[CLS + P] + SigLIP2 + WP | 97.56 $\pm$ 0.61 |

naive average pooling of patch tokens (+ AP) significantly degrades performance, indicating that uniform aggregation introduces background noise and weakens instance-discriminative cues. Combining pretrained global and local features partially recovers performance but still underperforms CLS-only embeddings. For the global features, adding a linear adaptation layer substantially improves performance, with SigLIP2 achieving the highest average accuracy among global representations. Weighted-average pooling consistently improves patch-based embeddings (e.g., DINOv2 [P] + WP improves from 72.77% to 96.54%), demonstrating the importance of selectively attending to informative regions. Combining local with global features or multiple models does not yield substantial gains over using global features alone, which already provide strong representations for effective instance-level discrimination. We further observe that training becomes more sensitive to weight initialization when combining tokens from different pretrained models or incorporating high-dimensional patch tokens. This suggests that a simple linear adaptation layer may be insufficient to effectively decode embeddings from distinct representation spaces or to fully leverage rich patch-level representations.

Overall, most methods achieve high accuracy (often exceeding 96%), indicating that the identification task may not be sufficiently challenging to clearly distinguish the discriminative capacity of different embeddings.

Table 3. Performance of image retrieval with three gallery image sets. All models except for the pretrained embeddings (\*) are trained with linear adaptation. Patch tokens are aggregated by weighted-average pooling (WP). Values are percentage. Bold text indicates the better performance comparing between w/wo patch tokens.

| Gallery Image Set            | Random Mobile Phone Image |             |              |             |             | Challenge Mobile Phone Image |             |              |             |             | Machine Image |             |              |             |             |
|------------------------------|---------------------------|-------------|--------------|-------------|-------------|------------------------------|-------------|--------------|-------------|-------------|---------------|-------------|--------------|-------------|-------------|
| Embeddings                   | MRR@k [%]                 |             | Recall@k [%] |             |             | MRR@k [%]                    |             | Recall@k [%] |             |             | MRR@k [%]     |             | Recall@k [%] |             |             |
|                              | k=10                      | k=100       | k=1          | k=10        | k=100       | k=10                         | k=100       | k=1          | k=10        | k=100       | k=10          | k=100       | k=1          | k=10        | k=100       |
| SigLIP2*                     | 44.2                      | 45.0        | 35.4         | 63.5        | 85.2        | 36.9                         | 37.9        | 28.1         | 57.1        | 81.1        | 36.9          | 37.8        | 28.5         | 55.8        | 76.5        |
| DINOv3[CLS]*                 | 49.8                      | 50.5        | 41.9         | 66.5        | 84.9        | 29.8                         | 30.6        | 23.6         | 44.4        | 66.2        | 29.8          | 30.6        | 23.6         | 44.4        | 66.2        |
| DINOv3[P] + WP               | 52.6                      | 53.5        | 41.7         | 76.6        | 95.6        | 46.8                         | 47.7        | 35.1         | 73.3        | 94.9        | 22.5          | 23.7        | 15.1         | 41.3        | 72.8        |
| SigLIP2                      | 57.6                      | 58.3        | 46.3         | 81.4        | 96.7        | 53.8                         | 54.6        | 42.3         | 78.6        | 96.8        | <b>37.4</b>   | <b>38.5</b> | <b>28.5</b>  | <b>59.6</b> | <b>87.3</b> |
| DINOv3[P] + SigLIP2 + WP     | <b>60.0</b>               | <b>60.7</b> | <b>48.8</b>  | <b>83.1</b> | <b>97.4</b> | <b>56.7</b>                  | <b>57.3</b> | <b>44.9</b>  | <b>81.6</b> | <b>97.3</b> | 30.5          | 31.6        | 21.3         | 53.0        | 81.0        |
| DINOv3 [CLS]                 | 53.5                      | 54.3        | 42.3         | 77.4        | 95.7        | 48.5                         | 49.5        | 37.0         | 74.0        | 95.2        | <b>25.0</b>   | <b>26.1</b> | <b>17.6</b>  | <b>43.6</b> | <b>72.7</b> |
| DINOv3[CLS + P] + WP         | <b>54.7</b>               | <b>55.5</b> | <b>43.5</b>  | <b>78.8</b> | 95.7        | <b>50.6</b>                  | <b>51.5</b> | <b>39.0</b>  | <b>76.2</b> | <b>95.4</b> | 22.4          | 23.6        | 15.4         | 39.7        | 69.2        |
| DINOv3[CLS] + SigLIP2        | 61.7                      | 62.3        | 50.3         | <b>87.4</b> | 97.8        | 57.8                         | 58.4        | 45.6         | 83.2        | 97.9        | <b>32.7</b>   | <b>33.8</b> | <b>23.3</b>  | <b>55.5</b> | <b>82.9</b> |
| DINOv3[CLS+P] + SigLIP2 + WP | <b>63.0</b>               | <b>63.6</b> | <b>51.6</b>  | 85.9        | <b>98.1</b> | <b>59.1</b>                  | <b>59.7</b> | <b>47.1</b>  | <b>84.0</b> | <b>98.0</b> | 29.7          | 30.9        | 20.8         | 51.6        | 80.2        |

## 5.2. Spare Part Image Retrieval

Beyond pairwise identification, large-scale retrieval requires selecting the correct match from thousands of candidates (e.g., 10,000), imposing much stronger demands on fine-grained discrimination. As described in Sec. 3.2.2, we evaluate three benchmarks and compare retrieval performance with and without patch tokens to assess the contribution of patch-level representations. The retrieval time is dominated by the encoder. For example, retrieval over a 10k gallery takes  $\sim 532$  ms per query for SigLIP2 + DINOv3 on an NVIDIA A100 (MIG 2g.20gb), with only  $\sim 50$  ms for linear adaptation and pooling and  $\sim 2$  ms for similarity search. We report results (Table 3) using the best validation-selected checkpoint obtained during training on the image identification task for each embedding under an identical training protocol. Since our primary goal is to fairly evaluate and compare the intrinsic representational capacity of different embeddings, we adopt a consistent model selection strategy that reflects their maximal achievable performance under the same adaptation setting, while minimizing the influence of optimization instability.

### 5.2.1. Effectiveness of CLS and Patch Tokens

Across most configurations, incorporating patch tokens consistently improves retrieval performance compared to CLS-only representations. Under the Random Mobile Phone gallery, patch-based variants outperform their CLS counterparts in nearly all metrics. For example, adding patch tokens to DINOv3[CLS] improves Recall@1 from 42.3% to 43.5%, with similar gains in MRR@10; when combined with SigLIP2, Recall@1 further increases from 50.3% to 51.6%. These improvements become more pronounced under the Challenge Mobile Phone gallery, where larger variations in viewpoint, background, and illumination make retrieval more difficult. As shown in 3, overall performance drops in this setting, but models with patch tokens

exhibit smaller degradation, indicating improved robustness to geometric and appearance changes.

This difference can be understood by comparing the roles of CLS and patch tokens. The CLS token encodes a global summary of the image and captures high-level semantic information, which is effective for identifying object categories and generalizing across poses. However, it may fail to capture fine-grained geometric details under challenging conditions such as occlusion, poor illumination, or background clutter. Moreover, CLS-based similarity often assigns similar scores to objects with comparable global structure, overlooking hard to visible but important shape differences (e.g., the last row of Figure 6).

In contrast, patch tokens provide localized representations that preserve fine-grained visual and geometric cues. This enables the model to better capture part-level correspondences and implicit 3D structure, which remain informative even when the viewing direction changes. As a result, patch-based retrieval is more robust to viewpoint variation and better at distinguishing structurally similar objects (e.g., the third row in Figure 6). However, relying solely on patch-level features also introduces limitations: the model may overemphasize local similarity and retrieve objects with similar parts but incorrect global configuration (e.g., the first row in Figure 6), indicating a weaker global structural prior. More examples are presented in Figure 6 and supplementary material.

Overall, these results suggest that patch tokens improve retrieval by encoding fine-grained structural and geometric information, leading to more robust performance under challenging conditions. However, their effectiveness is demonstrated within the same acquisition domain (mobile phone images), and may vary when generalizing to different domains, such as high-quality machine-captured images.



Figure 6. Top-3 retrieved images using DINOv3 CLS token and DINOv3 patch tokens + weighted-average pooling. Images with green frames indicate the correct matches, while reds are wrong retrieved objects.

### 5.2.2. Cross-Domain Generalization

Under the Machine Image gallery, this trend changes. In this cross-domain setting, patch-based variants often perform worse than their CLS-only counterparts. This suggests that patch-token aggregation is more sensitive to domain shift. A plausible explanation is that accurately weighting patch-level features requires domain-aligned training data. Since the weighted-average pooling layer is trained on mobile phone images, the learned patch weighting may overfit to domain-specific appearance patterns and fail to generalize to machine-captured images. In contrast, global CLS representations appear more robust to distribution shifts.

### 5.3. Re-ranking Using Local Presentation

To further leverage patch-level information, we experimented with re-ranking strategies using local representations. Specifically, we considered (1) Chamfer similarity [28] to align patch embeddings for fine-grained structural matching, and (2)  $\alpha$ QE [26] to incorporate neighborhood information from top-ranked gallery images. These methods aim to improve retrieval by emphasizing detailed spatial correspondence between query and gallery objects. Chamfer similarity directly compares local patch arrangements, while  $\alpha$ QE refines the query by aggregating features from high-confidence neighbors.

However, both strategies degrade overall retrieval performance. The primary cause is that patch-level representations may include background regions, which can introduce noise: patches corresponding to visually similar backgrounds may dominate the similarity computation, leading to incorrect rankings. These findings indicate that naive use of patch tokens for re-ranking can be detrimental. Proper

weighting or filtering of patch-level features, potentially learned in a domain-adaptive manner, is necessary to fully exploit local representations without introducing spurious similarity.

## 6. Conclusion & Future Work

We introduce a large-scale dataset for object retrieval in industrial spare parts, a challenging setting due to high visual similarity and subtle geometric differences between instances. The dataset provides a realistic benchmark with variations in viewpoint, background, and imaging conditions.

Using this dataset, we conduct an empirical study of pre-trained embeddings with simple and efficient adaptations. In particular, we analyze the roles of CLS and patch tokens, showing that they capture complementary information: CLS encodes global semantic cues, while patch tokens provide fine-grained geometric details. Rather than using patch tokens only for reranking, we directly incorporate them through weighted-average pooling, which brings consistent improvements in retrieval performance under challenging conditions. Overall, our study shows that combining global and local representations enables more accurate and fine-grained object retrieval, highlighting the potential of patch-level modeling for challenging visual recognition tasks.

Our current approach employs lightweight linear adaptation to learn patch importance. While efficient, this limits expressiveness. Future work will explore more powerful architectures, such as cross-attention between query and gallery images, to better model fine-grained spatial correspondences.



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